

Trace-Normalized Late-Memory Packets

A Gap-Free Dynamic Gramian and the Failure of Angle Perturbation

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Abstract

RH-85 showed that a clock-rank packet built from a strict-prefix snapshot can capture the terminal postblock state, while the unweighted prefix Gramian can retain large transient energy. This paper replaces the isolated snapshot by an online, scale-free late-memory statistic.

For $X_t = A^t S$ and $0 \leq \eta < 1$, define

$$G_j = \frac{X_j^* X_j}{\|X_j\|_2^2} + \eta G_{j-1}, \quad G_{-1} = 0.$$

The normalized-memory variational theorem identifies the leading rank- r projection of G_j as the minimizer of the exponentially weighted sum of relative snapshot residuals. The associated stacked tail bounds the current relative residual without any singular-value gap. RH-85 then propagates the packet through the unused suffix. This is a gap-free energy transfer, not a principal-angle argument.

At all five archived scales, one universal value $\eta = 1/512$ and packet time $\lceil 2M/3 \rceil$ give a direct 192-bit terminal residual below 1.10×10^{-5} and captured energy above 0.99999999988. Less than 0.196% of the normalized trace memory precedes the current snapshot. The weighted packet improves the raw unweighted prefix residual by at least three orders of magnitude and by more than 4.6×10^6 in the strongest case.

Conversely, every standard perturbation-to-gap ratio for comparing the memory packet to the point packet exceeds 8.9×10^6 , reaching above 8.1×10^9 . Thus the angle perturbation route is quantitatively unusable at the anchors even though captured energy is stable. Uniform weighted Rayleigh leakage, Stage A, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: dynamic Gramian; exponential forgetting; captured energy; principal angles; effective rank; interval arithmetic.

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1 A scale-free late-memory statistic

RH-84 isolated captured energy as the minimal effective-rank invariant [3]. RH-85 then proved snapshot transfer and found a strong strict-prefix packet, but also gave an explicit no-go theorem for unweighted prefix energy [4]. The problem is amplitude: a large early transient can dominate $\sum_{t \leq j} X_t^* X_t$ long after it has become irrelevant to the terminal state.

Normalize every nonzero snapshot by its Hilbert–Schmidt norm and forget the past geometrically. Define the stacked operator

$$\mathcal{N}_{\eta,j} u = \left(\eta^{(j-t)/2} \frac{X_t u}{\|X_t\|_2} \right)_{t=0}^j.$$

Its Gramian is exactly $G_j = \mathcal{N}_{\eta,j}^* \mathcal{N}_{\eta,j}$ and satisfies the online recursion in the abstract.

2 Gap-free variational and transfer theorems

Theorem 2.1 (Normalized-memory variational theorem). *Let $P_{\eta,j,r}$ be the leading rank- r spectral projection of G_j . Then*

$$\tau_r(\mathcal{N}_{\eta,j})^2 = \min_{\text{rank } P=r} \sum_{t=0}^j \eta^{j-t} \frac{\|X_t(I-P)\|_2^2}{\|X_t\|_2^2}, \quad (1)$$

$$\frac{\|X_j(I - P_{\eta,j,r})\|_2}{\|X_j\|_2} \leq \tau_r(\mathcal{N}_{\eta,j}). \quad (2)$$

No spectral gap is required.

Proof. The objective in (1) equals $\|\mathcal{N}_{\eta,j}(I-P)\|_2^2$. Ky Fan’s principle and Eckart–Young identify its minimizer with the leading spectral projection of G_j [1]. The $t = j$ summand has weight one and is bounded by the full nonnegative sum, proving (2). $\square$

Corollary 2.2 (Gap-free energy transfer). *For $j \leq M$,*

$$\frac{\|X_M(I - P_{\eta,j,r})\|_2}{\|X_M\|_2} \leq \frac{\|A^{M-j}\|}{\|X_M\|_2} \frac{\|X_j\|_2}{\|X_M\|_2} \tau_r(\mathcal{N}_{\eta,j}). \quad (3)$$

Proof. Apply the RH-85 snapshot transfer theorem to (2). $\square$

The total normalized trace memory is $(1 - \eta^{j+1})/(1 - \eta)$. Hence the mass before the current snapshot is at most $\eta/(1 - \eta)$. This deterministic bound prevents raw amplitude from overriding the intended time localization.

3 Why principal-angle control is the wrong gate

Write $\widehat{G}_j = X_j^* X_j / \|X_j\|_2^2$ and $E_j = G_j - \widehat{G}_j$. A Davis–Kahan route would compare spectral projections using $\|E_j\| / \delta_{j,r}$, where $\delta_{j,r} = \lambda_r(\widehat{G}_j) - \lambda_{r+1}(\widehat{G}_j)$ [2]. At a nearly finite-rank tail this gap can be much smaller than every physically relevant energy tolerance. The ratio may then be enormous although both packets capture essentially the same energy.

This does not contradict perturbation theory. It says only that identifying individual low-energy directions is much harder than bounding the total energy left outside a packet. The latter is exactly the invariant required by RH-84.

4 Five-scale 192-bit audit

We use $\eta = 1/512$, $j = \lceil 2M/3 \rceil$, and $r = \lceil H_\sigma \rceil + 2$. The packet is computed from the binary64 memory Gramian. Its snapshot and terminal residuals are then evaluated directly in 192-bit Arb arithmetic after exact binary lifting. The rank bound remains exact, and the Frobenius suffix inequality is interval certified.

The weighted packet is close to the point packet in captured energy, but not certifiably close in principal angle. The minimum gap ratio exceeds one million by a wide margin at every anchor. Meanwhile the normalized past trace fraction is below 0.001954 independently of the ambient dimension.

σ	rank	worst weighted tail	worst unweighted tail	min gap ratio
0.16	4	1.10×10^{-5}	1.38×10^{-1}	1.85×10^7
0.08	5	2.0×10^{-7}	1.10×10^{-1}	8.98×10^6
0.04	6	3.0×10^{-7}	2.45×10^{-1}	9.87×10^6
0.02	6	3.0×10^{-7}	2.84×10^{-1}	4.00×10^9
0.01	7	2.0×10^{-7}	3.26×10^{-1}	3.32×10^9

Table 1: Worst channel tails and the smaller directional gap ratio at each scale. Tail values are rounded conservatively from the generated audit.

5 Boundary and next theorem

The durable object is now the weighted Rayleigh leakage in (1). An all-level proof may bound it directly through a one-block energy recursion, without resolving unstable tail eigenspaces. A representative next target is a packet-energy drift inequality comparing $\text{tr}(P_j G_{j+1})$ with $\text{tr}(P_j G_j)$ plus a controlled injection term.

RH-86 proves the variational and transfer statements and validates the frozen late-memory packets. It does not derive η from the continuum dynamics, prove all-level weighted leakage, close Stage A1 or Stage A4, construct a relative fixed-disk determinant or self-adjoint Hilbert–Polya operator, identify zeta zeros, or prove the Riemann Hypothesis.

References

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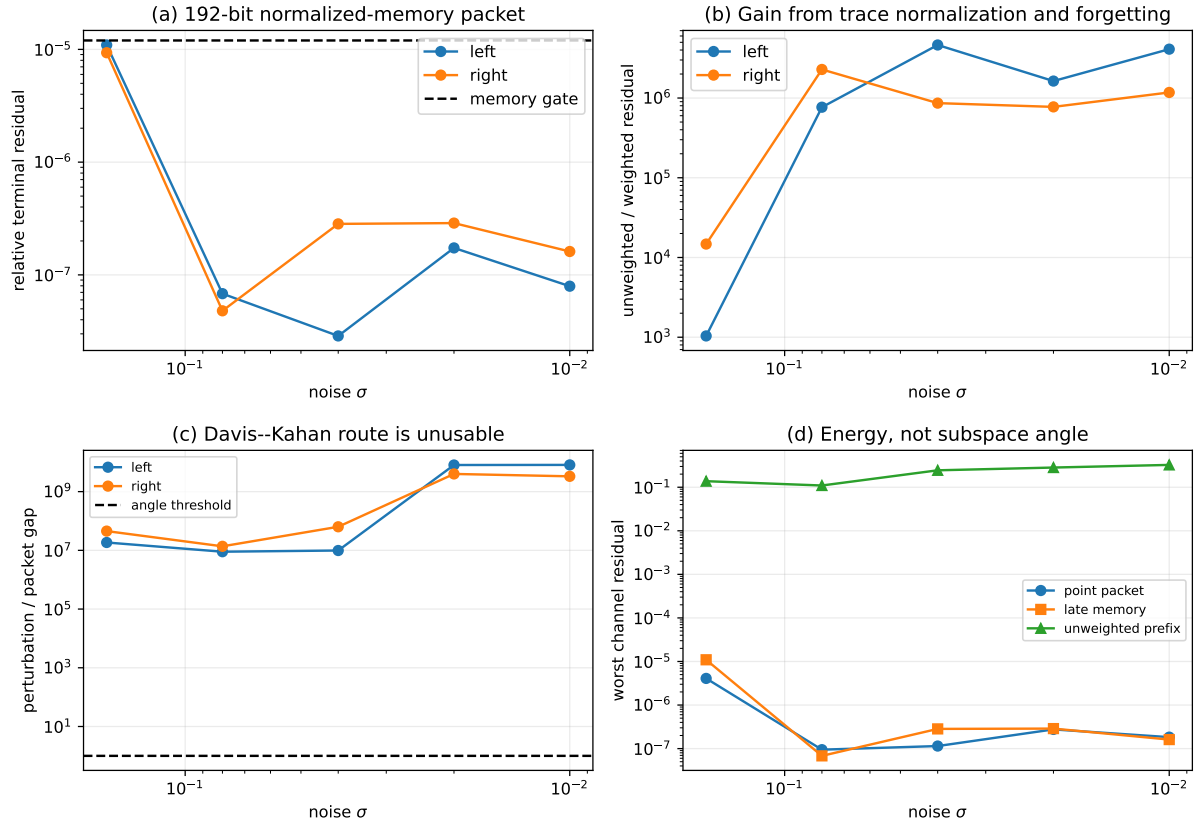


Figure 1: Interval late-memory residuals, improvement over raw prefix energy, the failure of gap-based angle control, and the three packet choices.